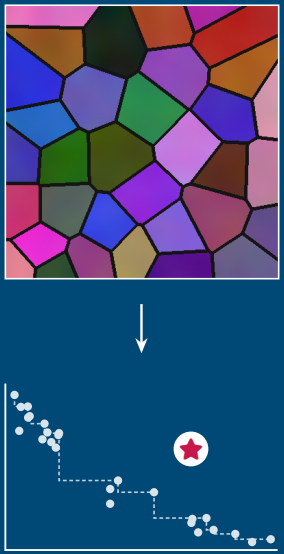
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Generative design of high-fidelity microstructures using physics-aware machine learning

Liao et al., Nat. Commun. 17, 11232 (2026)
https://doi.org/10.1038/s41467-026-78118-3

Camilo Martínez

d3M – Data-Driven Design of Materials
Chair of Experimental Methods in Materials Science
Universität des Saarlandes
Research group talk, 05.11.2026



Outline

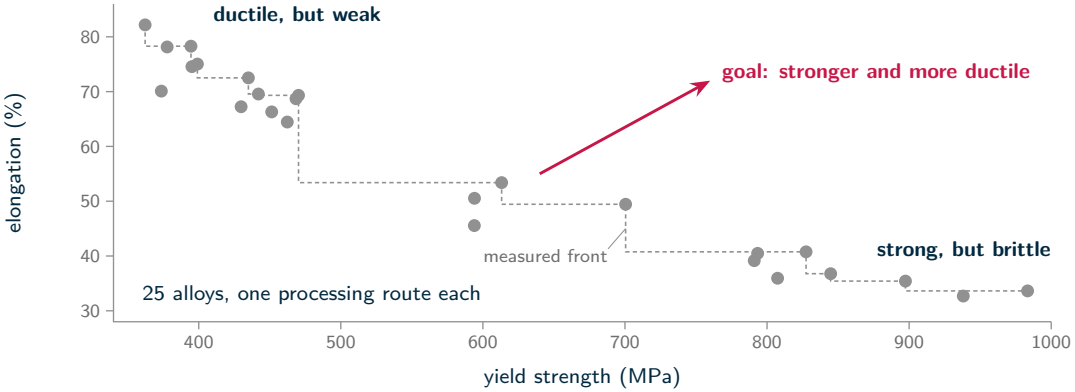
1	**The design problem**	Why ask a generative model for new microstructures?
2	**Groundwork**	Which methods does the paper build on?
3	**The method**	How are microstructures generated and steered toward properties?
4	**The results**	What do the authors show?

01

The design problem

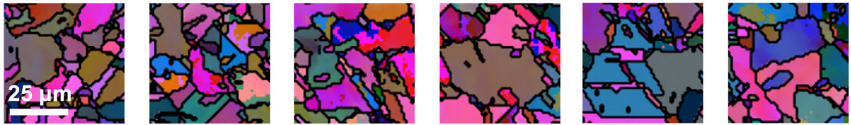
Why ask a generative model for new microstructures?

Strength/ductility trade-off



Classic pixelwise loss fails

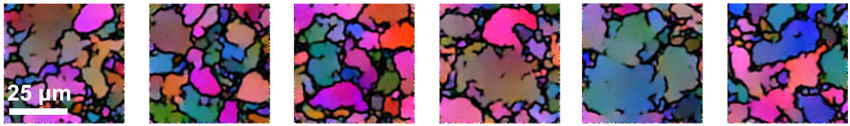
real patches



pixel loss only

$\text{MAE} = \frac{1}{n}\sum_i |x_i - \hat{x}_i|$

$\text{MSE} = \frac{1}{n}\sum_i (x_i - \hat{x}_i)^2$

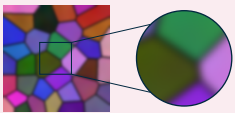


blurred or missing grain boundaries, colours mixed inside grains

Two fixes

the obstacle

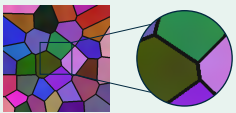
$L = L_{\text{pix}}$



more terms

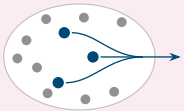
the paper's fix

$L = L_{\text{pix}} + L_{\text{edge}} + L_{\text{SSIM}}$



sharp boundaries, one colour per grain

random sampling



new codes fall among the training codes

steer by properties

property-guided search



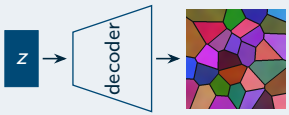
02

Groundwork

Which methods does the paper build on?

Three ways to generate an image

In one pass



a decoder turns a code into an image

VAE (2013), GAN (2014)

whole image at once

Token by token



a network predicts each tile from the tiles before it

DALL-E 1 (2021); language models write text this way

no going back to refine

Step by step from noise



a network removes a little noise at each step

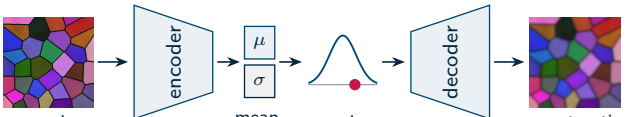
DDPM (2020), Stable Diffusion (2022), DALL-E 2 (2022)

refines the whole image

Liao et al. join the first and the third: the VAE output guides the diffusion steps

Kingma and Welling (2013), arXiv:1312.6114 · Goodfellow et al. (2014), NeurIPS · Ramesh et al. (2021), ICML · Ho et al. (2020), NeurIPS · Rombach et al. (2022), CVPR

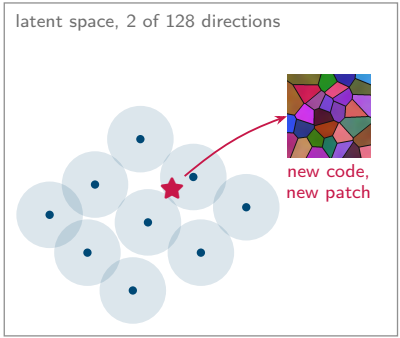
Variational Autoencoder (VAE)



$$L = L_{\text{rec}} + L_{\text{KL}}$$

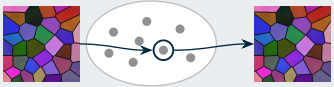
output close to the input

codes close to a standard normal



Reconstruct, generate, extrapolate

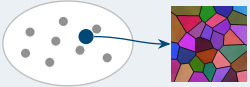
Reconstruct



a training patch, encoded and decoded

compare with its input

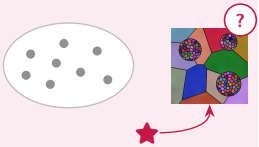
Generate



a new code among the training codes

compare with held-out data

Extrapolate



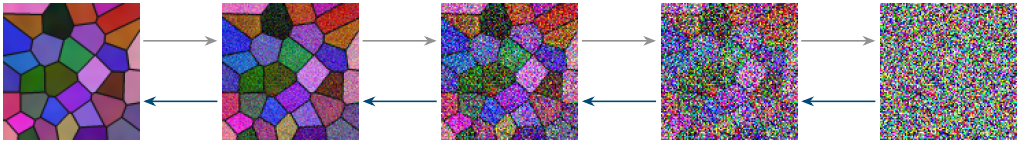
a code away from all training codes

only an experiment can tell

Diffusion models

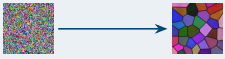
forward: at each of 1000 steps, mix in a little noise (fixed)

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\,\epsilon$$

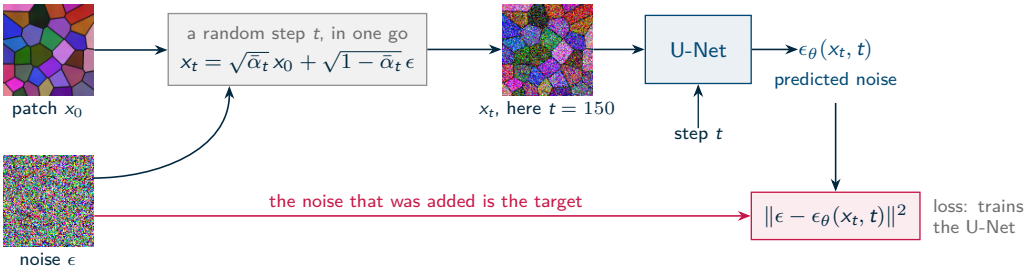


reverse: a U-Net predicts the noise, removed step by step (learned)

generating with the trained model, two ways:

unconditional		from noise alone, step by step: a new patch of any kind
conditional		the U-Net also sees a guide at every step, here the VAE output

Training a diffusion model

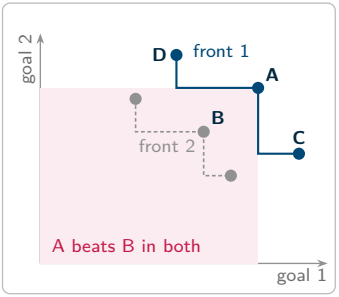


repeated over many patches, steps and noise images: one U-Net learns all 1000 steps

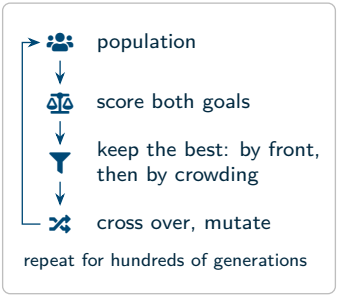
Liao et al.: the U-Net also sees the VAE output, $\epsilon_\theta(x_t, t, x_V)$

Pareto fronts and NSGA-II

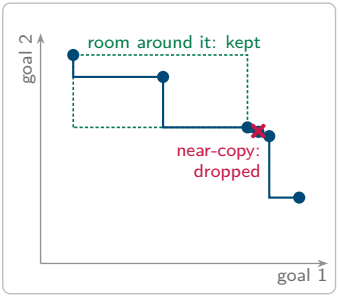
NS
non-dominated sorting



GA
genetic algorithm



II
crowding distance



Pareto (1906), *Manuale di economia politica* · Holland (1975), *Adaptation in Natural and Artificial Systems* · Deb et al. (2002), IEEE Trans. Evol. Comput. 6, 182



03

The method

How are microstructures generated and steered toward properties?

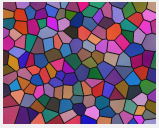
25 alloys, one map each

hot-rolling reduction at 900 °C

annealing temperature	0 %	20 %	30 %	40 %	50 %
650 °C	1 h	20 h	15 h	10 h	5 h
750 °C	5 h	1 h	10 h	20 h	15 h
850 °C	10 h	5 h	20 h	15 h	1 h
950 °C	15 h	10 h	1 h	5 h	20 h
1050 °C	20 h	15 h	5 h	1 h	10 h

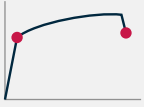
each cell one alloy: its annealing time; darker: higher yield strength

Each alloy



EBSD map
1000×800 px
0.5 µm per px

colour: $(\varphi_1, \Phi, \varphi_2) \rightarrow$ red, green, blue



tensile test: yield strength, elongation

Alloy split before cropping

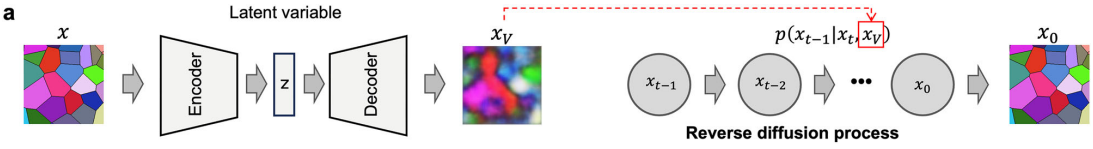
EBSD maps | patches, 100×100 px | + flips, rotations | + degraded copies



		×80		×6		×6	
training	**20 alloys**	→	**1,600**	→	**9,600**	→	**57,600**
test	**5 alloys**	→	**400**	→	**2,400**	→	**14,400**

✔ test patches come from five alloys the model never sees in training

A VAE, refined by diffusion



variational autoencoder (VAE)
three convolutional layers in, three transposed out
loss $L_{\text{pix}} + L_{\text{edge}} + L_{\text{SSIM}} + L_{\text{KL}}$

conditional diffusion model (DDPM)
a U-Net predicts the noise, guided by x_V
loss: the same three terms on the noise

training data
57,600 images of the 20 training alloys: patches to reproduce, and degraded copies to restore

generation
a code from the search goes through the decoder, then the diffusion model: a new patch

Optimization Loss

$$L = L_{\text{pix}} + L_{\text{edge}} + L_{\text{SSIM}}$$

pixel values

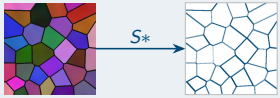
$$\frac{1}{n}\sum_{i=1}^{n}|x_i - \hat{x}_i|$$

Mean Absolute Difference (MAE)

every pixel weighs the same; boundaries are few pixels

edges

$$\text{mean}\,|S * x - S * \hat{x}|$$



Sobel filter S, across and down: large at grain boundaries

structure

$$1 - \text{SSIM}(x, \hat{x})$$

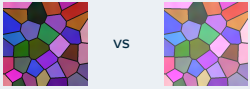


means, contrasts and correlation: orientation inside grains

Liao et al. (2026), Nat. Commun. 17, 11232, Eqs. 1–9 · Sobel and Feldman (1968), Stanford AI Project · Wang et al. (2004), IEEE Trans. Image Process. 13, 600

Structural Similarity (SSIM)

luminance L

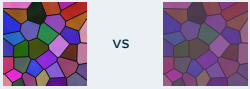


$L = 0.89 \quad C = 1.00 \quad S = 1.00$

$$\frac{2\mu_x\mu_{\hat{x}} + C_1}{\mu_x^2 + \mu_{\hat{x}}^2 + C_1}$$

brightness: the means μ

contrast C

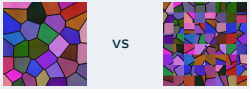


$L = 1.00 \quad C = 0.70 \quad S = 1.00$

$$\frac{2\sigma_x\sigma_{\hat{x}} + C_2}{\sigma_x^2 + \sigma_{\hat{x}}^2 + C_2}$$

contrast: the spreads σ

structure S



$L = 1.00 \quad C = 1.00 \quad S = 0.02$

$$\frac{\sigma_{x\hat{x}} + C_3}{\sigma_x\sigma_{\hat{x}} + C_3}$$

pattern: the covariance $\sigma_{x\hat{x}}$

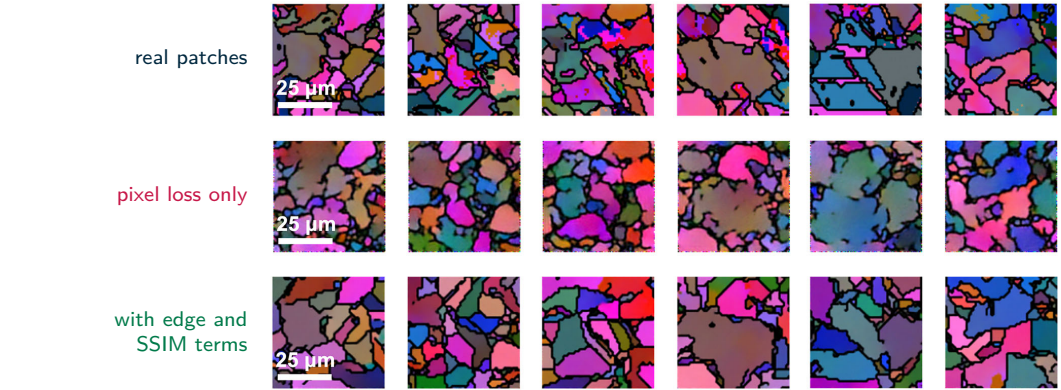
$$\mathrm{SSIM} = L^\alpha\, C^\beta\, S^\gamma$$

paper: $\alpha = \beta = \gamma = 1$, $C_3 = C_2/2$

$$= \frac{(2\mu_x\mu_{\hat{x}} + C_1)(2\sigma_{x\hat{x}} + C_2)}{(\mu_x^2 + \mu_{\hat{x}}^2 + C_1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + C_2)} \longrightarrow L_{\mathrm{SSIM}} = 1 - \mathrm{SSIM}$$

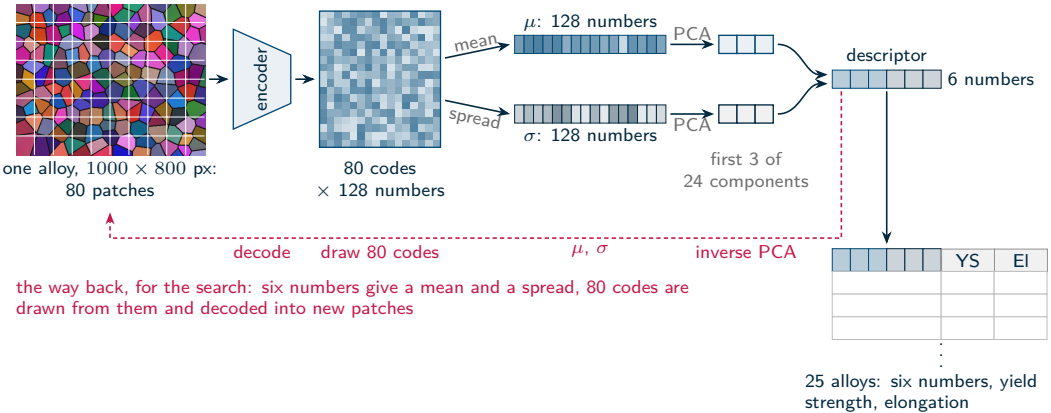
zero for identical patches

The same patches, with the new loss



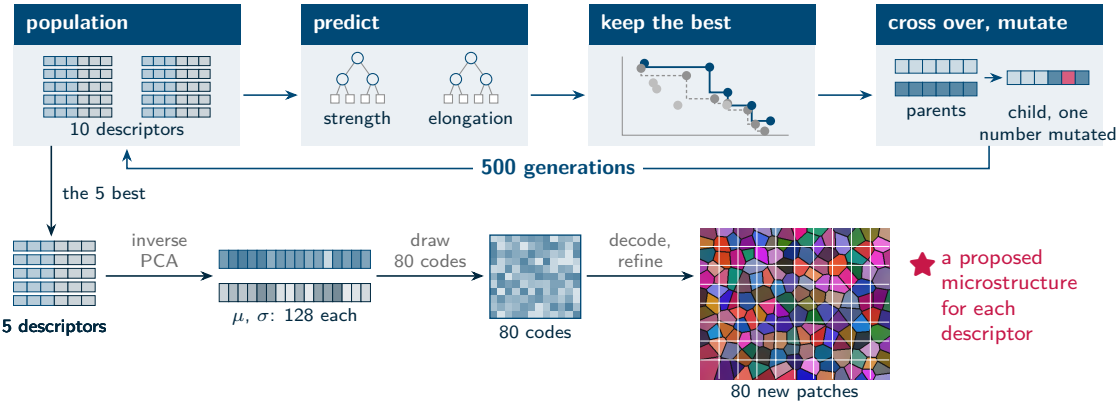
Liao et al. (2026), Nat. Commun. 17, 11232, Fig. 2c (CC BY-NC-ND 4.0)

Six numbers per alloy



the way back, for the search: six numbers give a mean and a spread, 80 codes are drawn from them and decoded into new patches

A genetic search in six numbers

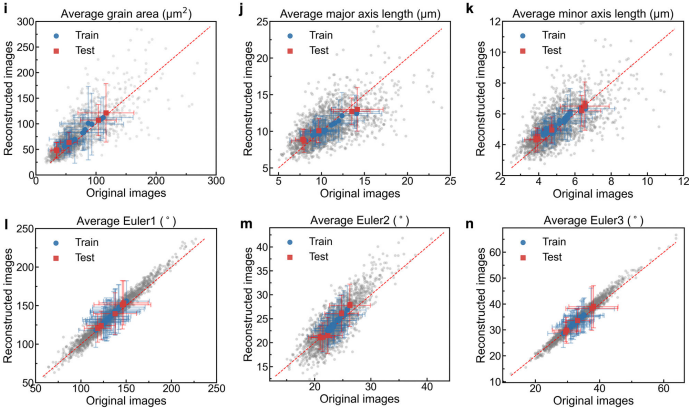


04

The results

What do the authors show?

Grain statistics after reconstruction



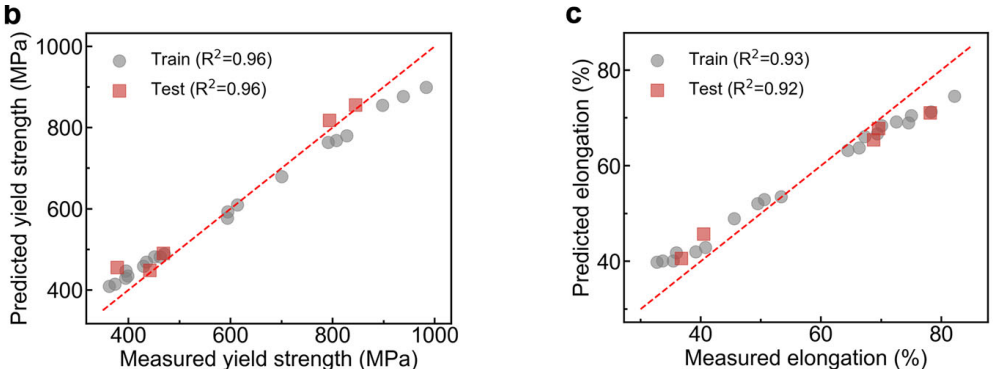
original against reconstruction

i–k grain area and ellipse axes

l–n the three Euler angles

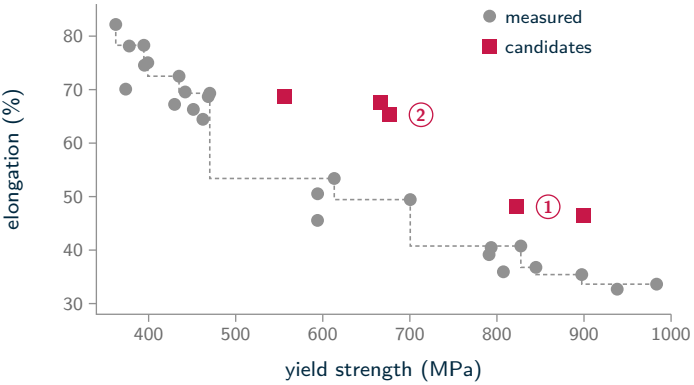
- 2,000 single patches: wide scatter
- 20 training alloys, each the mean of 80 patches
- 5 test alloys
- reconstruction = original

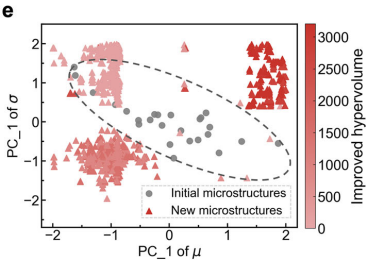
Property predictions



Gradient-Boosted Trees on the six numbers: fitted on 20 alloys, tested on 5

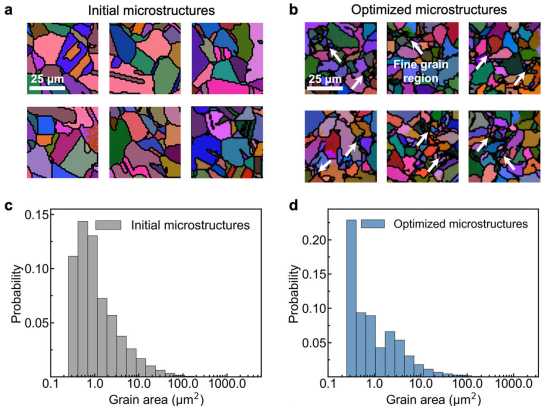
Five candidates beyond the front





first principal components of mean and spread: generated (red) beside measured (grey)

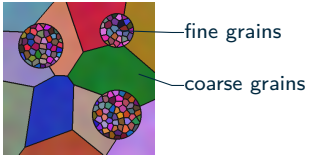
Fine grains in the generated patches



b fine-grain regions, marked by arrows

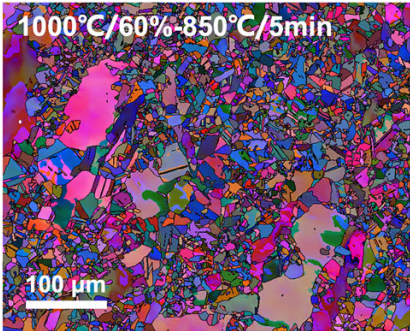
d more grains in the smallest bin

proposed mechanism

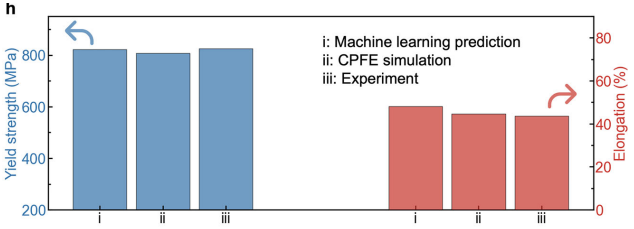


- fine grains block dislocations: strength
- coarse grains let them glide: ductility
- the mismatch stores dislocations: hardening

Simulation and one experiment



new alloy, route from the literature: 60 % rolling at 1000 °C, 5 min at 850 °C



candidate ①: i predicted, ii simulated by crystal plasticity (CPFE), iii measured on the new alloy



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Thank you

Camilo Martínez

A1 The loss terms in full

Sobel kernels and edge term (Eqs. 2, 3)

$$S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad S_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$L_{\text{edge}} = \text{mean}\big(|S_x * x - S_x * \hat{x}| + |S_y * x - S_y * \hat{x}|\big)$$

structural similarity (Eqs. 8, 9)

$$\text{SSIM}(x, \hat{x}) = \frac{(2\mu_x\mu_{\hat{x}} + C_1)(2\sigma_{x\hat{x}} + C_2)}{(\mu_x^2 + \mu_{\hat{x}}^2 + C_1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + C_2)}$$

$$L_{\text{SSIM}} = 1 - \text{SSIM}(x, \hat{x})$$

VAE, with x_V the reconstruction of x (Eq. 11)

$$L_{\text{VAE}} = L_{\text{pix}} + L_{\text{edge}} + L_{\text{SSIM}} + L_{\text{KL}}\big(q(z|x), p(z)\big)$$

diffusion: forward process (Eq. 13)

$$q(x_t|x) = \mathcal{N}\big(x_t; \sqrt{\bar{\alpha}_t}\, x, (1 - \bar{\alpha}_t)\mathbf{I}\big)$$

diffusion: loss on the predicted noise $\epsilon_\theta(x_t, t, x_V)$ (Eq. 15)

$$L_{\text{DDPM}} = L_{\text{pix}}(\epsilon_\theta, \tilde{\epsilon}) + L_{\text{SSIM}}(\epsilon_\theta, \tilde{\epsilon}) + L_{\text{edge}}(\epsilon_\theta, \tilde{\epsilon})$$

A2 The 25 processing routes

ID	annealing °C	annealing h	rolling %	measured MPa	measured %
1	650	1	0	470	69.3
2	750	5	0	468	68.7
3	850	10	0	435	72.5
4	950	15	0	395	78.3
5	1050	20	0	362	82.2
6	1050	15	20	378	78.2
7	950	10	20	399	75.0
8	850	5	20	613	53.4
9	750	1	20	700	49.4
10	650	20	20	827	40.8
11	650	15	30	897	35.4
12	750	10	30	793	40.5
13	850	20	30	594	50.5

ID	annealing °C	annealing h	rolling %	measured MPa	measured %
14	950	1	30	442	69.6
15	1050	5	30	395	74.6
16	1050	1	40	430	67.2
17	950	5	40	451	66.3
18	850	15	40	594	45.5
19	750	20	40	791	39.1
20	650	10	40	983	33.6
21	650	5	50	938	32.7
22	750	15	50	845	36.8
23	850	1	50	807	35.9
24	950	20	50	462	64.5
25	1050	10	50	374	70.1

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The paper and its code

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